Executive Briefing & Business Insights

Global Crop Yield & Agricultural Input Analytics

101 countries · 10 crops · 1990–2013

Descriptive and diagnostic analytics

1. Headline Figures

Metric	Value	Note
Observations analyzed	13,130	Country-crop-year records
Average yield	70,969 hg/ha	Pooled across all 10 crops — see Section 3 for per-crop figures
Year-over-year yield growth	1.23%	Average annual growth, 1990–2013, gap-aware across the missing 2003 data point
Countries covered	101	
Average pesticide use	14,838.70 tonnes	Country-year average (see Section 4)
Yield per pesticide tonne	4.78	Efficiency ratio
Yield vs. temperature correlation	-0.14	Weak negative association
Yield vs. rainfall correlation	0.01	Effectively no linear association
Top-yielding crop	Potatoes	182,602 hg/ha average
Top-producing nation	United Kingdom	By average yield

2. What the Data Shows

- Global average yield rose from roughly 62,000 hg/ha in 1990 to roughly 83,000 hg/ha by 2013 — a sustained upward trend across the full period.
- The top seven yielding nations by average (United Kingdom, Belgium, Denmark, Netherlands, Ireland, New Zealand, Sweden) are all high-latitude, temperate-climate producers. This reflects yield density (hg/ha), not total production volume.
- Correlation between yield and either temperature or rainfall is weak (-0.14 and 0.01 respectively). Within this dataset, temperature and rainfall alone explain very little of the variation in yield density — crop choice, farming practice, and other factors plausibly dominate.

3. Yield by Crop Type

Roots and tubers are physically bulky crops weighed by fresh mass; cereals and legumes are weighed dry, by grain only. This difference in natural physical scale is why individual crop averages differ from the pooled headline figure by up to 10x:

Crop	Avg. yield (hg/ha)	Category
Potatoes	182,602	Root / tuber
Plantains and others	110,838	Root / tuber
Sweet potatoes	104,317	Root / tuber
Yams	103,759	Root / tuber
Cassava	102,705	Root / tuber
Rice, paddy	37,542	Cereal

Crop	Avg. yield (hg/ha)	Category
Maize	36,980	Cereal
Wheat	30,783	Cereal
Sorghum	17,995	Cereal
Soybeans	16,564	Legume

Reading guidance: the 70,969 hg/ha headline figure sits between these two clusters because it pools all ten crops together. Read it as "average across everything grown," not as a representative figure for any single crop.

4. Interpreting the Pesticide Figure

Average pesticide use is reported as 14,838.70 tonnes per country-year. This is intentionally an average rather than a total: pesticide use is recorded once per country-year and repeats across every crop grown that year, so a total summed across crop rows would overstate real-world use many times over. The average per country-year is the figure that carries a genuine, sanity-checkable interpretation.

5. Scope and Framing

This dashboard answers what happened, where, and how strongly these factors are associated — it does not forecast what will happen next. No regression, forecasting, or machine-learning component exists in this model. A future extension toward prediction would need its own train/test methodology and reported error metrics, and should be presented as a separate, clearly labeled addition.